



HOUSE PRICE PREDICTION

Predicting Residential Property Prices with Machine Learning

Data Science / Machine Learning Project

Regression Analysis • Feature Engineering • Model Evaluation



Problem Statement & Objective

Real estate prices depend on a complex mix of location, size, quality, and market factors. Manual valuation is slow and inconsistent — this project builds a data-driven model to estimate a home's sale price from its measurable attributes.



Objective

Build and compare regression models that accurately predict house sale prices, and identify which features drive value the most.



What drives price?

Identify the features most correlated with sale price.



Which model fits best?

Compare linear, tree-based, and ensemble regressors.



How accurate?

Evaluate using RMSE, MAE, and R^2 on unseen data.

Dataset Overview

2,930

Property Records

79

Features / Attributes

23

Neighborhoods

1

Target: Sale Price

Key Feature Categories



Location

Neighborhood, zoning, lot config



Size

Living area, lot area, basement sq. ft.



Layout

Bedrooms, bathrooms, room count



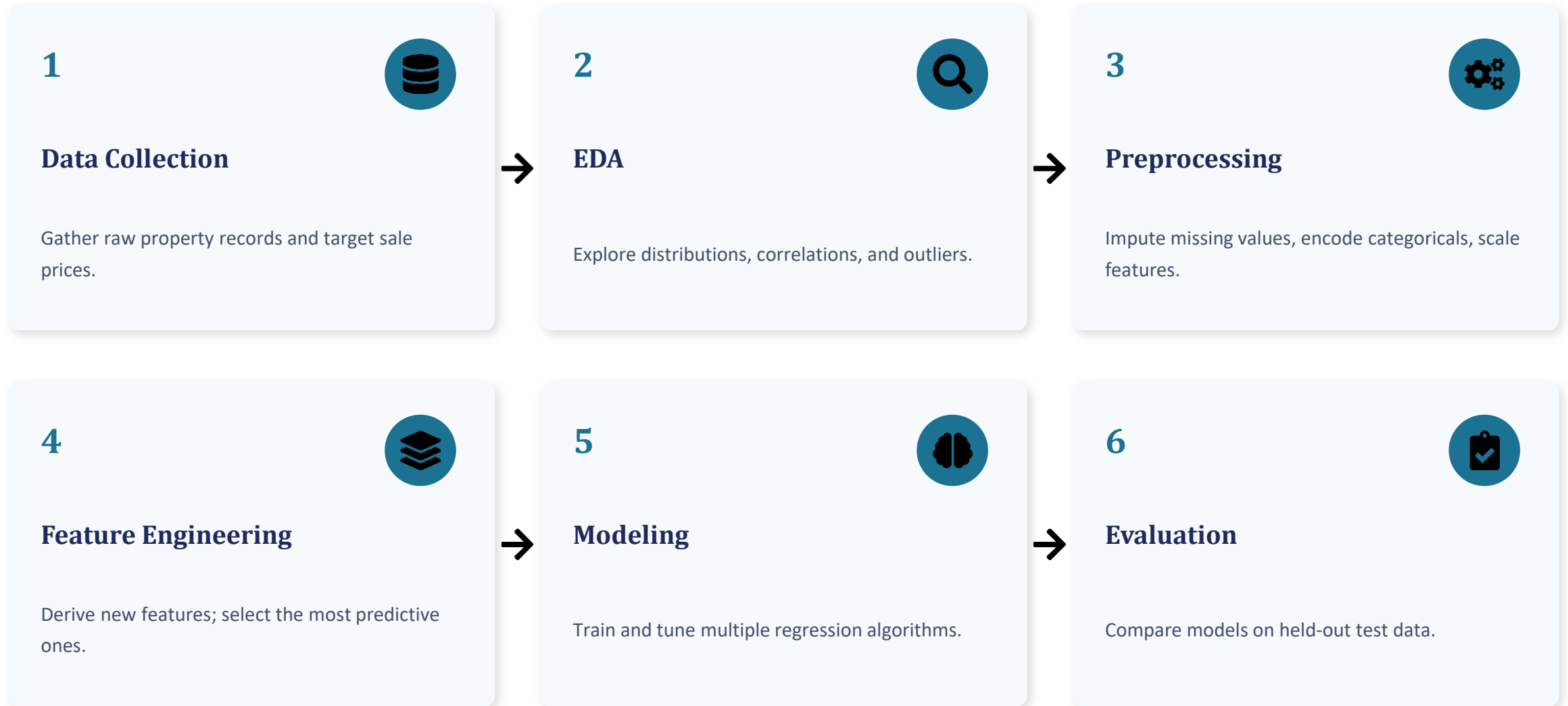
Quality

Overall quality, condition, year built



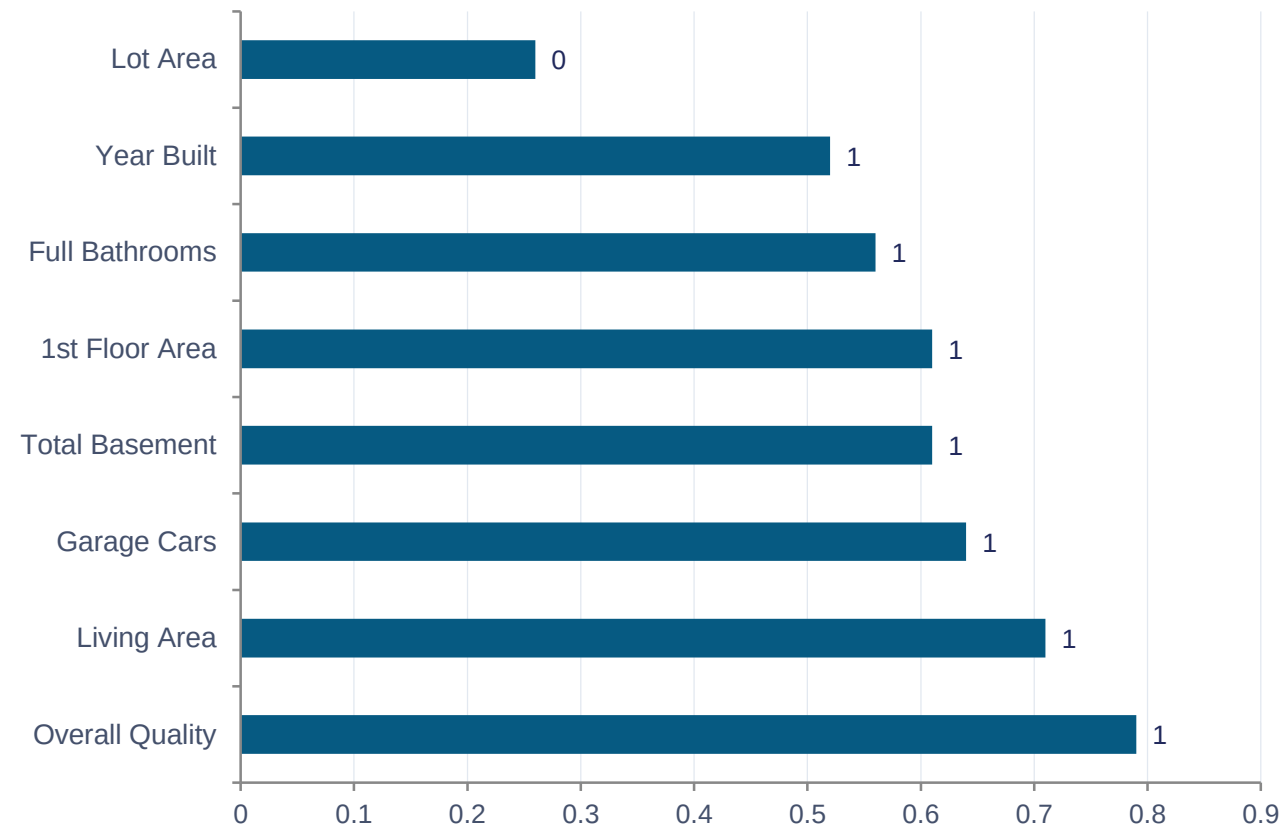
Data quality: missing values in features like lot frontage and garage year; outliers in living area and sale price were treated before modeling.

Methodology & Pipeline



Exploratory Data Analysis

Correlation with Sale Price



Key Insights

- Overall quality rating is the single strongest predictor of price.
- Living area and basement size show strong positive correlation.
- Sale price is right-skewed — a log transform improves model fit.
- Neighborhood location creates distinct price clusters.

Feature Engineering

Raw attributes were transformed into features that better expose the relationships a model can learn from.



Total Living Area

Combine basement + 1st + 2nd floor square footage into one measure of total space.



House Age at Sale

Derived from sale year minus year built — captures depreciation and renovation effects.



Total Bathrooms

Full and half baths across all levels combined into a single comparable count.



Quality × Area Interaction

Combines overall quality with size, since large-but-low-quality homes price differently.



One-Hot Encoded Categoricals

Neighborhood, zoning, and style converted into model-ready binary indicators.



Log-Transformed Target

Sale price log-transformed to reduce skew and stabilize variance for linear models.

Models Evaluated



Linear Regression

Baseline model assuming a linear relationship between features and price.



Random Forest

Ensemble of decision trees that captures non-linear feature interactions.



Gradient Boosting (XGBoost)

Sequentially built trees that correct prior errors — typically the strongest performer.

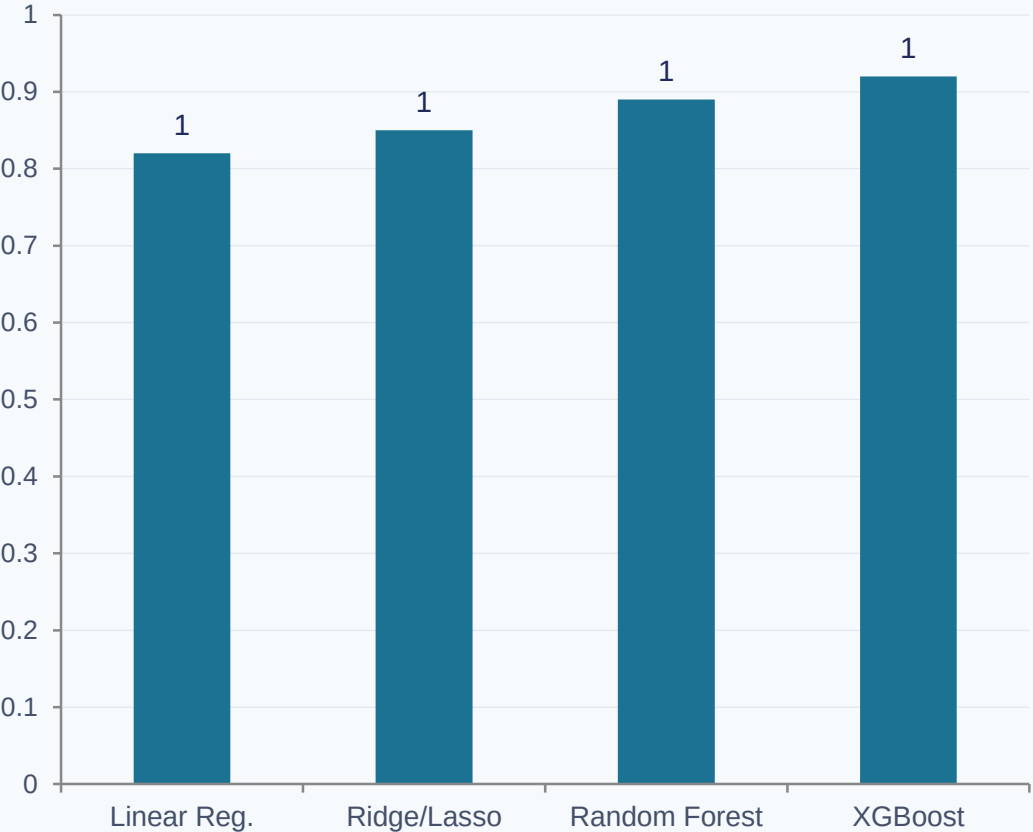


Ridge / Lasso Regression

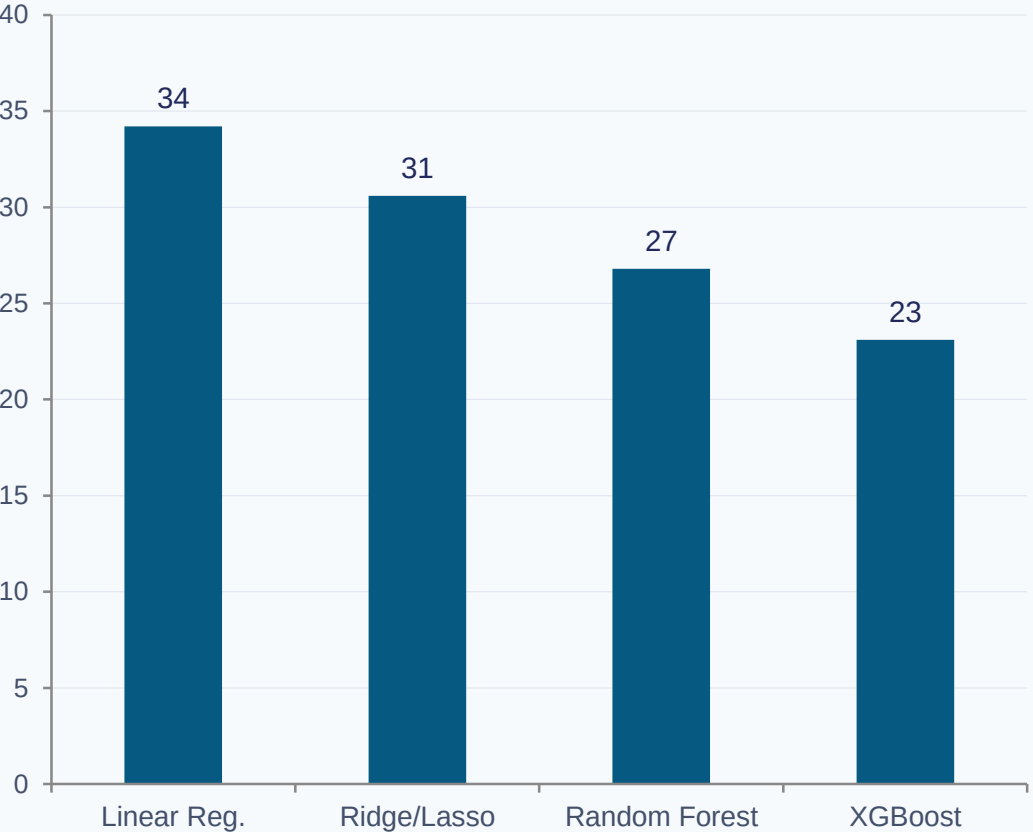
Regularized linear models that reduce overfitting on high-dimensional data.

Model Performance Comparison

R² Score (higher is better)

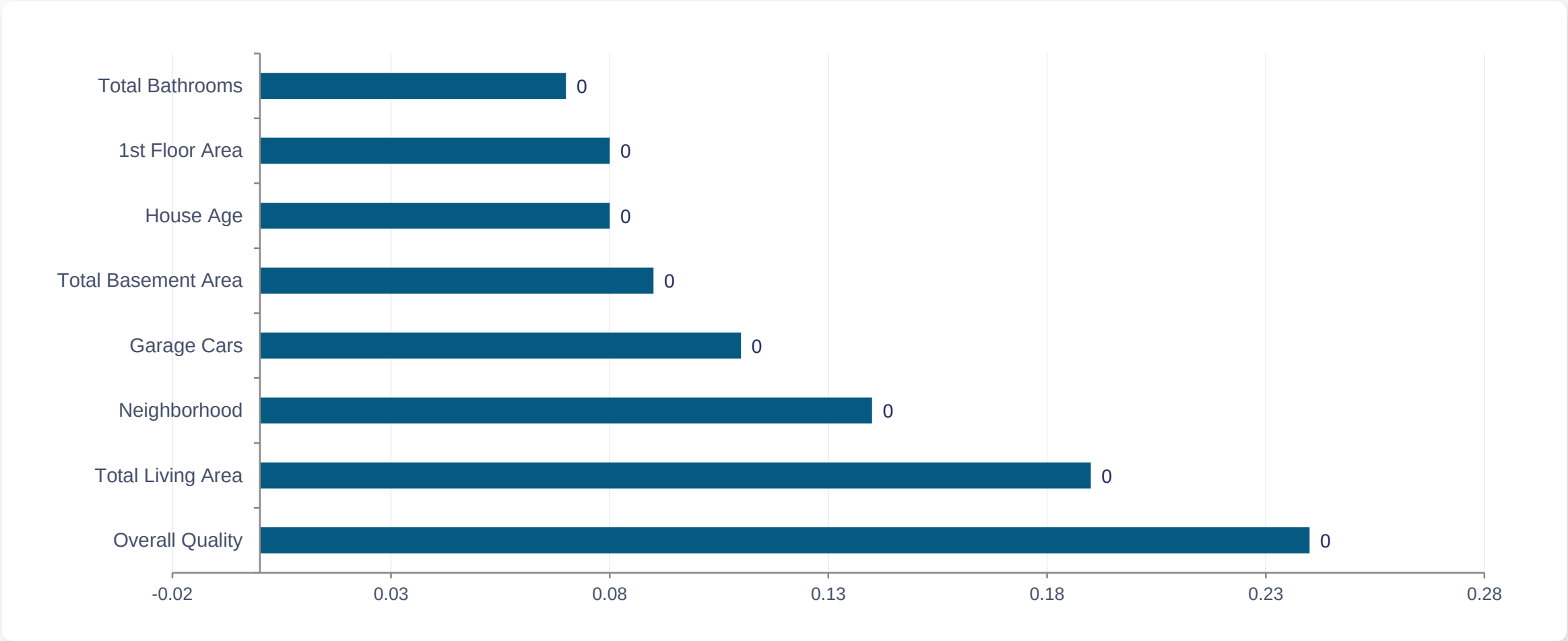


RMSE, \$K (lower is better)



What Drives the Prediction?

Feature importance from the best-performing model (XGBoost)



Conclusion & Future Work



Key Takeaways

- Gradient boosting (XGBoost) delivered the best accuracy, explaining ~92% of price variance.
- Overall quality and living area are the dominant price drivers.
- Feature engineering — combining areas, encoding location — measurably improved every model.
- The pipeline generalizes: it can be retrained on new listings with minimal changes.



Future Work

- Incorporate external data: school ratings, crime rates, walkability scores.
- Try stacked ensembles and neural network regressors for further gains.
- Add SHAP-based explanations for per-listing price justification.
- Deploy as a live web app for real-time price estimates.